

Can word ranks reveal benchmark exposure?

Two experiments compare an OpenWebText-trained GPT-2 with MiniCPM5 Base. Both use exact next-word top-1, top-5 and top-10 accuracy: first against familiar and novel controls, then across OLMES benchmark dates.

Research question

If a model encountered benchmark questions during training, it might predict their exact wording unusually well. We test whether OLMES receives an advantage beyond what general language ability produces, and whether the newer model loses that advantage on later benchmark families.

Predict the question, not the answer

Answer options and correct answers are excluded. Each model receives the first half of a text and predicts the words in the remaining half. Scoring is teacher forced: before every prediction, the model sees all preceding true words. For BoolQ and SocialQA, the text contains context followed by the question, so some predicted words are context.

Outcomes: exact next-word top-1, top-5 and top-10 accuracy. A target is a hit when its complete space-delimited string, including attached punctuation, is among the model's first k candidates within the shared vocabulary described next. Higher accuracy means easier wording prediction; this is not question-answering accuracy.

Why use word ranks?

Raw negative log probability is sensitive to calibration. Two models can order candidates identically while assigning different probability values because one has sharper logits. Top-k accuracy depends only on order, so additive logit shifts and positive temperature rescaling cannot change it.

The models use different subword vocabularies. We therefore intersected their canonical one-token word strings. The shared candidate set contains **28,853 strings**, each consisting of one leading ASCII space followed by a segment with no whitespace. The same strings are ranked for both models.

A target is evaluated only when both tokenizers encode it as the corresponding single canonical token. Multi-token words are excluded from both models rather than counted as misses. Eligible positions and denominators match exactly. This covers 80.2% of OLMES pilot continuation words.

Why report three cutoffs? Top-1 asks whether the true word is the model's first choice. Top-5 and top-10 allow plausible alternatives. Agreement across cutoffs would be stronger evidence than selecting whichever k looks favorable.

Tradeoff: rank removes probability calibration and candidate-tokenizer differences, but it ignores the size of probability gaps and excludes multi-token words. It also does not remove differences in intelligence, model size, domain knowledge or training volume.

Models

Older: Stanford GPT-2 Small, 124M parameters, documented as trained on OpenWebText. Newer: MiniCPM5-1B-Base, 1.08B parameters, whose mixture includes Ultra-FineWeb and other datasets. Both pinned base checkpoints predate the IOL 2026 control. They are not a controlled architecture pair.

Experiment 1 / Design and controls

Question: does OLMES wording receive an unusually large word-prediction advantage relative to familiar corpus excerpts and genuinely recent human-written text?

OLMES: 30 items, three from each of ARC Easy, ARC Challenge, BoolQ, CommonsenseQA, HellaSwag, OpenBookQA, PIQA, SocialIQA, WinoGrande and MMLU.

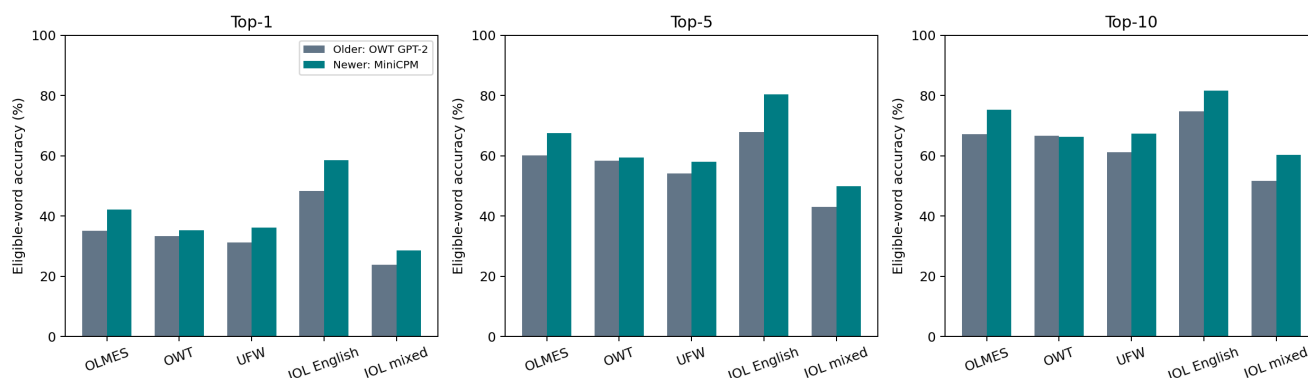
Corpus controls: 30 OpenWebText and 30 Ultra-FineWeb excerpts, roughly matched to OLMES length. They are verified corpus members, but exact consumption by the selected checkpoints is unverified and they are not genre matched to exam questions.

Post-checkpoint control: 30 non-overlapping mixed-text excerpts from five human-composed IOL 2026 problems. Because unfamiliar languages and tables make those excerpts a poor format match, ten short English-prose excerpts from the same five problems form the main sensitivity control.

The rank experiment covers 130 natural-language texts. Fresh SHA-256 strings from an earlier diagnostic are omitted because each is one space-delimited segment with no next-word position. IOL has five independent parent-problem clusters, not 30 independent questions.

Experiment 1 / Control results

Exact next-word prediction within the shared one-token vocabulary



OLMES: GPT-2 scores 35.1%, 60.1% and 67.2% at top-1, top-5 and top-10. MiniCPM scores 42.2%, 67.6% and 75.3%. Its gains are +7.1, +7.5 and +8.1 percentage points; paired cluster-bootstrap intervals exclude zero in this pilot.

Relative to each model's own corpus control, the OLMES advantage rises by 4.3 top-1 points, 7.9 top-5 points and 7.4 top-10 points. This is consistent with extra familiarity, but the corpus controls are not genre matched.

Crucial sensitivity: on novel English IOL, MiniCPM gains +10.3 points at top-1 and +12.6 at top-5—larger than its OLMES gains. General predictive ability can therefore produce at least as much improvement as the proposed familiarity signal.

Mixed IOL has only 52.2% shared-word coverage and is difficult for both models; English IOL has 72.5% coverage and is the more informative control. Experiment 1 does not isolate direct exposure.

Experiment 2 / Hypothesis and design

We next stay within OLMES and ask whether MiniCPM’s relative word-prediction advantage shrinks as benchmark date increases.

Specific prediction: if the newer model benefits disproportionately from exposure to older benchmark wording and that benefit fades for later families, newer-minus-older accuracy should have a negative date slope.

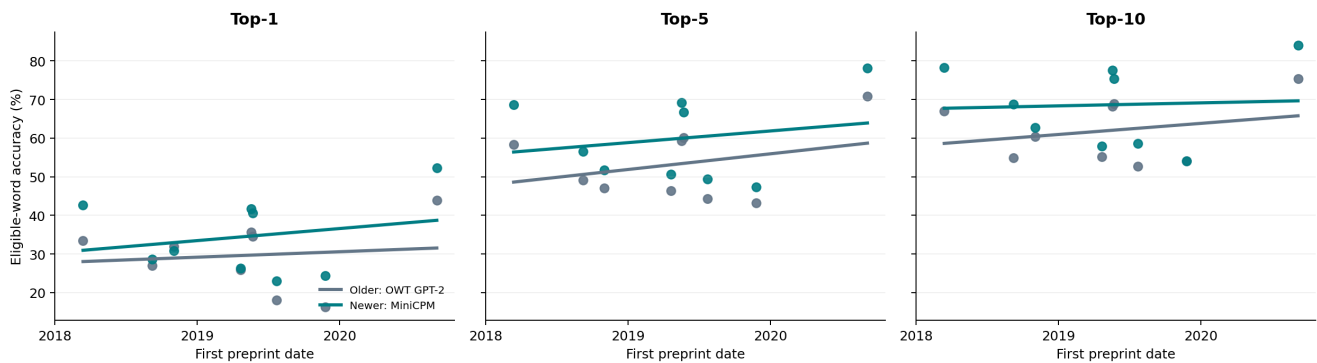
We froze a separate sample of 300 items, 30 per component. There are 296 paired texts with at least one eligible continuation word and **3,842 identical eligible positions per model**.

Dates are first arXiv submission dates from 2018–2020. They are benchmark-age proxies, not verified creation dates for individual questions. ARC Easy and Challenge share one paper and are averaged into one family, leaving nine equally weighted families. MMLU is the only 2020 family.

Expected if the hypothesis is true: MiniCPM’s advantage falls as date increases. A downward fitted line in the direct advantage plot supports that direction. Because every family predates MiniCPM, leakage need not produce this pattern.

Experiment 2 / Accuracy by benchmark date

Question-word prediction by benchmark date

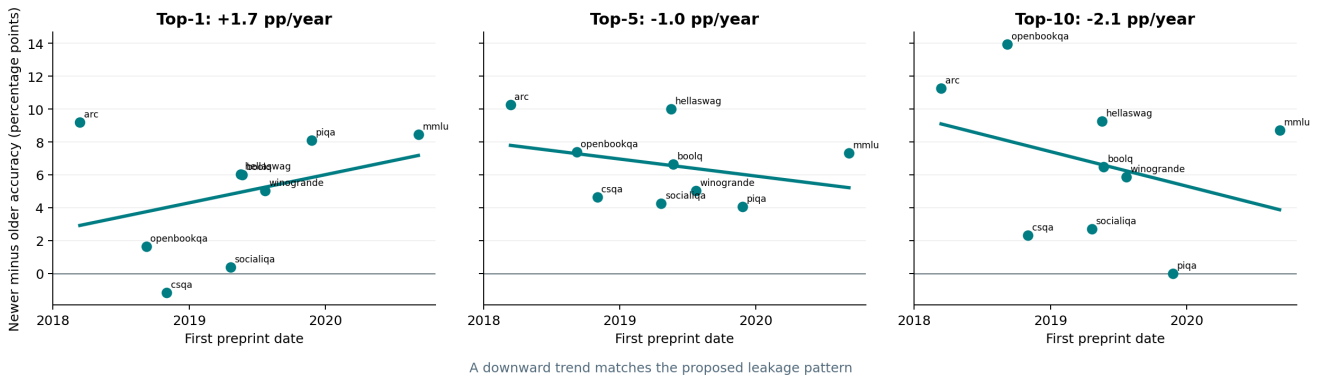


Both models' fitted accuracy rises with date, but date is entangled with task composition. At top-1, GPT-2 rises +1.4 points/year and MiniCPM +3.1. At top-5 the slopes are +4.1 and +3.0; at top-10, +2.9 and +0.8.

These individual slopes do not isolate familiarity. The direct test subtracts GPT-2 accuracy from MiniCPM accuracy within each family, shown next.

Experiment 2 / Direct advantage slopes

Does the newer model lose its word-prediction advantage?



Observed: newer-minus-older accuracy slopes +1.7 points/year at top-1, -1.0 at top-5 and -2.1 at top-10. Top-1 points opposite the prediction; top-5 and top-10 point with it.

The descriptive 95% family-bootstrap intervals are [-2.9, +7.9] for top-1, [-3.6, +1.3] for top-5 and [-9.0, +2.9] for top-10 percentage points/year. Every interval includes zero. The direction changes with k , and nine heterogeneous families provide limited precision.

Experiment 2 therefore does not detect the proposed temporal degradation. An interval crossing zero is not proof of equal slopes or absence of exposure.

Interpretation and reproducibility

Experiment 1: OLMES gets a meaningful newer-model rank gain, but the novel English control gets a larger top-1 and top-5 gain. **Experiment 2:** top-1 and top-5/top-10 disagree on temporal direction, and all intervals cross zero.

Word ranks solve the cross-model calibration objection. They do not solve the intelligence confound. These results do not provide convincing evidence that leakage explains the model difference, and they do not establish that either model is contamination free.

Stronger follow-up: compare multiple years of the same exam series across a verified cutoff, match subject and format, and include confirmed consumed records. Extending the method to exact multi-token word probabilities would improve coverage without reverting to raw cross-model calibration.

Execution and saved evidence

FP16 inference ran locally on a GTX 1650 (4 GB). The final control and temporal rank passes took 107.9 and 103.4 seconds. Cost: \$0. Five project scoring tests passed. Invariants verified canonical target IDs, identical eligible counts and $\text{top-1} \leq \text{top-5} \leq \text{top-10}$.

Saved outputs include pilot group summaries, cluster contrasts, paired model advantages, temporal component means, family-bootstrap statistics and the shared-vocabulary definition and hash. The public bundle excludes sampled source text and model weights.

Pinned GPT-2: 2f4e2079c9ac92c2b5c6fecc19fae645bcef56fa. Pinned MiniCPM: f120ef901e3a11c2ed5fcfd242296cea3e80191b. OLMES commit: 5a51f502d463b8cdc4a2dcad7d7096c41ff1197e.

Sources: [Patel and Han](#); [Stanford checkpoint documentation](#); [MiniCPM checkpoint](#); [OLMES](#); [IOL 2026 problems](#).